

PHISHING AWARENESS TRAINING

CodeAlpha Cyber Security Internship
Task 2: Phishing Awareness Training

Prepared By:
Aditya Sai Teja Ravilisetti

This presentation aims to educate users about phishing attacks, social engineering tactics, and cybersecurity best practices. It provides practical guidance on identifying phishing emails, fake websites, and protecting sensitive information from cybercriminals.

What is Phishing?

Phishing is a type of cyberattack in which attackers impersonate legitimate organizations, businesses, or individuals to trick victims into revealing sensitive information. The goal of phishing is usually to steal usernames, passwords, banking details, credit card information, or personal data.

Attackers often use fake emails, websites, text messages, or phone calls that appear trustworthy. Victims are encouraged to click malicious links, download infected attachments, or provide confidential information.

Why Phishing is Dangerous

- Easy to execute
- Difficult for many users to identify
- Can result in financial loss
- Can lead to identity theft
- May compromise organizational data

Statistics

According to cybersecurity reports, phishing remains one of the most common causes of data breaches worldwide.



TYPES OF PHISHING ATTACKS

Common Types of Phishing

1. Email Phishing

Attackers send fraudulent emails pretending to be banks, social media platforms, online stores, or government agencies.

2. Spear Phishing

Targeted attacks aimed at specific individuals or organizations. Attackers gather personal information to make emails appear genuine.

3. Smishing

Phishing conducted through SMS messages containing malicious links or fake alerts.

4. Vishing

Voice phishing using phone calls to convince victims to reveal confidential information.

5. Clone Phishing

Attackers duplicate legitimate emails and replace safe links with malicious ones.

6. Whaling

A highly targeted attack aimed at executives, CEOs, and senior management personnel.



HOW PHISHING ATTACKS WORK

Phishing Attack Lifecycle

Step 1: Preparation

The attacker creates a fake email, website, or message designed to look legitimate.

Step 2: Delivery

The phishing message is sent to potential victims through email, SMS, social media, or messaging apps.

Step 3: Interaction

The victim clicks a malicious link or opens an attachment.

Step 4: Data Collection

The victim unknowingly enters confidential information on a fake website.

Step 5: Exploitation

The attacker uses the stolen information for fraud, unauthorized access, or identity theft.

Consequences

- Financial loss
- Data theft
- System compromise
- Reputation damage



HOW TO RECOGNIZE PHISHING EMAILS

Warning Signs of Phishing Emails:-

Suspicious Sender Address

Cybercriminals often use email addresses that closely resemble legitimate organizations.

Example:

✗ support@amaz0n-security.com

✓ support@amazon.com

Urgent or Threatening Language

Messages often create panic:

- "Your account will be suspended."
- "Immediate action required."

Grammar and Spelling Mistakes

Many phishing emails contain unusual wording or grammatical errors.

Suspicious Links

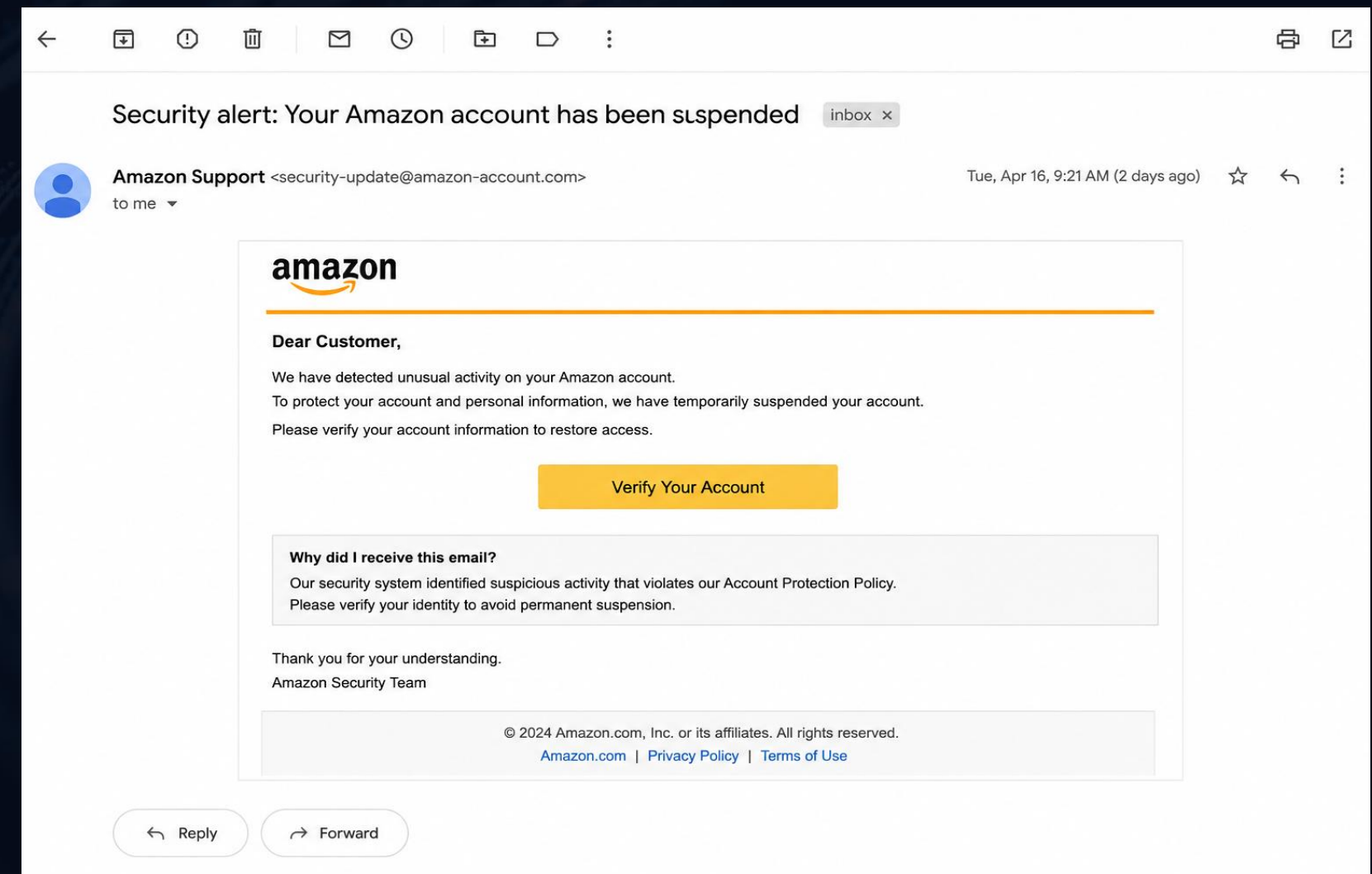
Always hover over a link before clicking.

Unexpected Attachments

Attachments may contain malware designed to infect your device.

Requests for Sensitive Information

Legitimate organizations rarely ask for passwords or banking information through email.



IDENTIFYING FAKE WEBSITES

How to Detect Fake Websites:-

Check the URL

Attackers often create websites with slightly modified domain names.

Example:

Real Website:

<https://www.amazon.com>

Fake Website:

<https://www.amaz0n-login.com>

Verify HTTPS

Secure websites use HTTPS and display a padlock icon.

Check the Security Certificate

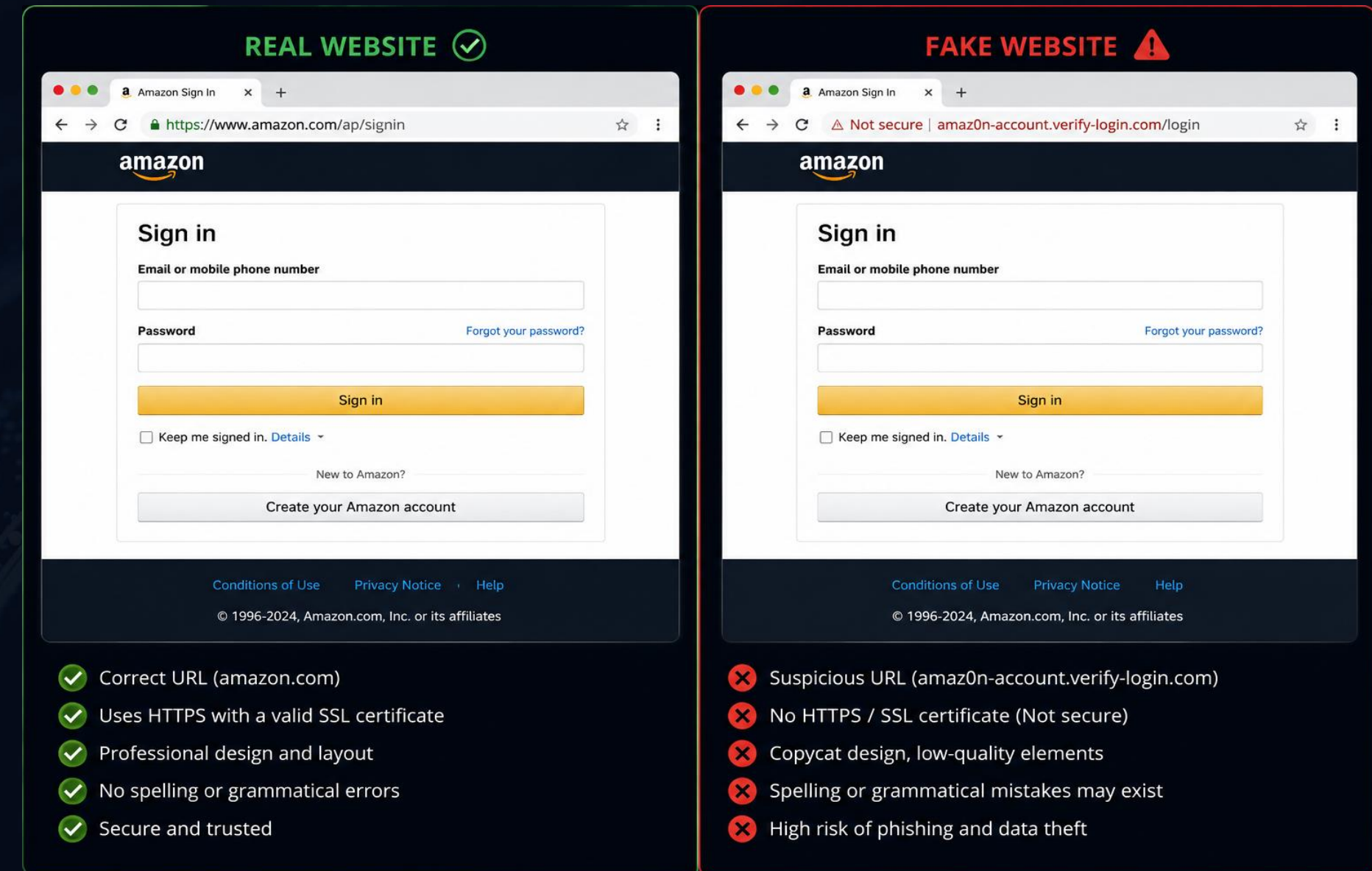
View the site's certificate information to verify authenticity.

Inspect Website Design

Poor quality graphics, broken pages, and inconsistent layouts may indicate a fake website.

Avoid Sharing Credentials

Never enter usernames, passwords, or banking information unless you are certain the website is legitimate



SOCIAL ENGINEERING TACTICS USED BY ATTACKERS

Understanding Social Engineering

Social engineering involves manipulating human emotions and behavior to gain unauthorized access to information.

Fear

Example:

"Your account has been compromised."

Urgency

Example:

"Respond within 24 hours to avoid account closure."

Authority

Example:

"This message is from your IT department."

Curiosity

Example:

"See who viewed your profile."

Rewards

Example:

"You have won a free smartphone."

Why It Works

People often react emotionally rather than logically, making them vulnerable to manipulation.



REAL-WORLD PHISHING EXAMPLE

Banking Phishing Scam:-

A user receives an email appearing to come from a trusted bank.

The email states:

"Due to suspicious activity, your account has been temporarily locked. Click the link below to verify your identity."

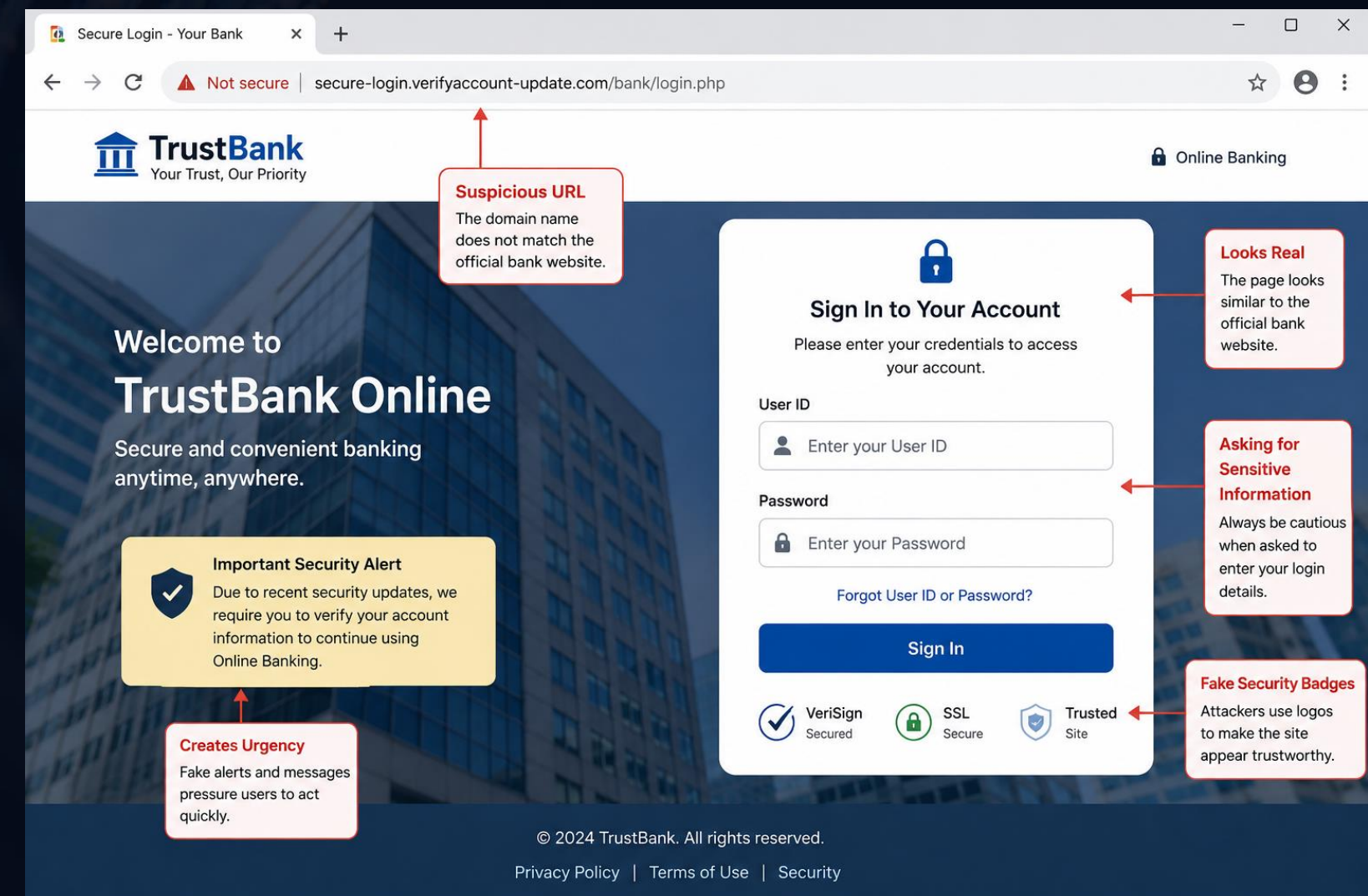
The user clicks the link and is redirected to a fake banking website that looks identical to the real one. After entering login credentials, the information is sent directly to the attacker.

Impact:-

- Financial fraud
- Unauthorized transactions
- Identity theft
- Data compromise

Lesson Learned:-

Always verify emails and websites before entering sensitive information.



BEST PRACTICES TO AVOID PHISHING ATTACKS

How to Stay Safe:-

Verify the Sender

Check email addresses carefully before responding.

Hover Over Links

Inspect URLs before clicking.

Use Multi-Factor Authentication (MFA)

Adds an extra layer of security.

Use Strong Passwords

Create unique and complex passwords.

Keep Software Updated

Updates often include security patches.

Report Suspicious Messages

Inform your IT department or service provider.

Never Share Sensitive Information

Legitimate organizations rarely ask for passwords via email.

Think Before You Click

Take a moment to evaluate any unexpected message.

CYBERSECURITY BEST PRACTICES

Simple steps for a safer digital life



QUIZ AND CONCLUSION

Interactive Quiz

Question 1

Which of the following is a common sign of phishing?

- A. Secure HTTPS connection
- B. Official company email
- C. Urgent request for credentials ☒
- D. Verified website certificate

Question 2

What should you do before clicking a link?

- A. Open immediately
- B. Verify the URL by hovering over it ☒
- C. Ignore security warnings
- D. Share it with friends

Question 3

What is Spear Phishing?

- A. SMS-based phishing
- B. Targeted phishing attack against a specific individual ☒
- C. Voice phishing
- D. Malware attack



Conclusion:-

Phishing attacks continue to be one of the most common cybersecurity threats worldwide. By understanding phishing techniques, recognizing warning signs, and following cybersecurity best practices, individuals and organizations can significantly reduce their risk of becoming victims.

Impact of Phishing Attacks:-

- Financial Loss
- Identity Theft
- Data Breaches
- Reputation Damage
- Malware Infection
- Business Disruption

Final Message:-

"Think Before You Click!"
Stay Alert • Stay Safe • Stay Secure